# Amir Mirzai Golpayegani


Calgary, AB, Canada

amirgolpaa24@gmail.com | amir.mirzaigolpayega@ucalgary.ca | +1 (587) 439-0436

linkedin.com/in/amir-golpayegani-218148165 | github.com/amirgolpaa24 | Portfolio


## Summary


M.Sc. Computer Science graduate with experience building Python/C++ systems for machine learning, NLP-adjacent backend services, model evaluation, data pipelines, and large-scale data processing. Skilled in PyTorch, Scikit-learn, REST APIs, SQL, Docker, Git, Linux, and reproducible experimentation, with applied research experience in retrieval, data infrastructure, LLM workflows, and AI automation.


## Education

**M.Sc. in Computer Science** GPA: **3.94**/4.0 Sep 2022 – Feb 2026
*University of Calgary* *Calgary, Canada*

**B.Sc. in Computer Engineering** GPA: **3.69**/4.0 Sep 2016 – Feb 2021
*Sharif University of Technology* *Tehran, Iran*

## Technical Skills

**Programming:** Python, C++, JavaScript, SQL
**ML / AI:** PyTorch, Scikit-learn, TensorFlow, CNNs, model training, evaluation, LLM workflows, AI agents
**Data / NLP:** NumPy, Pandas, SciPy, text processing, retrieval workflows, Google Earth Engine API, GDAL, Rasterio
**Backend / Engineering:** Django, Node.js, REST APIs, PostgreSQL, Docker, Git, Linux, AWS, GitHub Actions
**Visualization / Automation:** Streamlit, Polyscope, OpenGL, GLSL, n8n, OpenClaw, SearXNG, Telegram API

## Work Experience

**Research Software Developer** Sep 2022 – Feb 2026
*BigGeo / Mitacs, University of Calgary* *Calgary, Canada*

- Built C++/Python software for scalable processing, storage, and retrieval of large Earth observation datasets.
- Automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs.
- Developed DGGS-based encoding pipelines to convert satellite imagery into structured tensor files.
- Improved encoding performance with C++ multithreading, reducing 10-tensor processing from 233s to 26s.
- Reduced storage by 38% for scaled Canada-wide data with 1,112 GeoTIFF tiles and 16K+ DGGS tensors.

**Back-End Developer** Jun 2020 – Jul 2022
*Asr Gooyesh Pardaz* *Tehran, Iran*

- Developed Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL.
- Built authentication features including registration, Google login, password reset, and email/phone verification.
- Implemented backend logic for payment-service integration and storing plan/payment records in a relational database.
- Contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data to answer client questions.

**Deep Learning Teaching Assistant** Jul 2026
*Neuromatch Academy* *Remote, United States*

- Mentored an international pod of ∼15 students through Neuromatch Academy's intensive three-week Deep Learning course.
- Facilitated hands-on PyTorch tutorials spanning MLPs, CNNs, attention/transformers, diffusion generative models, and reinforcement learning.
- Guided project groups through end-to-end deep learning research projects, from question formulation and literature review to model implementation and final presentations.

**Teaching Assistant** Dec 2022 – Apr 2025
*Department of Computer Science, University of Calgary* *Calgary, Canada*

- **Developing Big Data Applications**: Mentored +25 students on AWS-based data pipelines, cloud storage, debugging, and scalable data workflows.
- **Working with Data at Scale**: Supported +20 students with large-scale data processing, data pipelines, and high-volume data analysis.
- **Database Management Systems**: Assisted +75 students with SQL, database design, course projects, grading, and exam support.

## Research and Technical Projects

**Agentic Job Search Automation Platform** May 2026 – Present
*OpenClaw, n8n, LLMs, AI Agents, Docker, Workflow Automation, SearXNG, Telegram API*

- Building an agentic AI system to search jobs, analyze job fit, and generate tailored application materials.
- Designed LLM-based workflows for resume/job matching, skill-gap analysis, and application decision support.

- Integrated Google Sheets, Google Drive, and Telegram API to track jobs and send automated application alerts.
- Prototyped the initial workflow in n8n before migrating toward a more flexible OpenClaw-based agent system.

**DEM Super-Resolution Framework** May 2024 – Feb 2026

*Python, PyTorch, CNNs, Scientific Imaging, Google Earth Engine API, Polyscope, Data Processing and Visualization*

- Trained deep ResNet models for $4\times$ DEM super-resolution, reconstructing 30m elevation data from 120m inputs.
- Built a Streamlit pipeline using Rasterio to parallelize GEE DEM tile downloads and collect mountainous terrain data worldwide.
- Implemented PyTorch training and evaluation workflows with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors.
- Analyzed model generalization across geographic regions and used evaluation results to compare reconstruction quality.

**Spatiotemporal Earth Observation Data System Using DGGS** Sep 2022 – Feb 2026

*First-author publication* | *M.Sc. thesis* | *C++, Python, DGGS, Wavelets, B-splines, Time-Series*

- Designed a DGGS-based framework for real-time multiresolution management of satellite imagery.
- Developed a tensor-based storage model for direct access to arbitrary regions at multiple resolutions.
- Built B-spline temporal approximation to compress time-series data and retrieve missing timestamps locally.
- Retrieved 105,489 DGGS cells per time-series query in 13.9s, compared with about 220s using global retrieval.

## Publication

**Amir Mirzai Golpayegani**, Mahmudul Hasan, Faramarz F. Samavati. *Real-Time Multiresolution Management of Spatiotemporal Earth Observation Data Using DGGS*. Remote Sensing, 2025. doi:10.3390/rs17040570

## Certificates

**Supervised Machine Learning: Regression and Classification**, DeepLearning.AI and Stanford University, Coursera Oct 2024

**Applied Machine Learning in Python**, University of Michigan, Coursera Jul 2020

## Languages

**English:** Full Professional Proficiency (TOEFL **115**/120)

**Persian:** Native Proficiency

## Achievement

**Teaching Assistant Excellence Laureate** 2024/2025

Department of Computer Science, University of Calgary — awarded for outstanding teaching support (*link*)

**Ranked 17th Nationwide in Iran University Entrance Exam (Konkour)** May 2016

Top 0.001% among all participants